

1) What is Simple linear regression? State how to use Simple linear regression with its relevant formula? (5 marks)

→ Simple linear regression is a statistical method used to understand the relationship between two variables

- i) One dependent Variable (the outcome or response)
- ii) One independent Variable (prediction or explanatory Variable)

In Simple, terms it predicts the value of dependent Variable based on the value of independent Variable

Formula for Simple linear regression

General formula for Simple linear regression is

$$Y = mx + c$$

where Y is the dependent variable (the variable we want to predict)

x is the independent Variable (the variable we use to make prediction)

c is the intercept (the value of Y when $x=0$)

m is the slope of regression line
(change in Y for one unit of change in x)

How to use Simple linear regression

① Collect Data: Gather data for both dependent and independent Variable. For example, if you want to predict house prices (Y) based on area (Sq.foot), collect historical data on both variables

② Plot the data: Create a scatter plot to visualize the relationship between independent Variable (x) and dependent variable (Y)

3) Calculate the regression line

• The slope (m) can be calculated using formula

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

• The intercept c can be calculated as using formula

$$Y = mx + c$$

$$\therefore \boxed{c = \bar{y} - m\bar{x}}$$

where $\bar{x}$ and $\bar{y}$ are means of the X and Y variables

4) plot the regression line

Draw the regression line using the equation

$Y = mx + c$. This line represents best fit for data points, minimizing the sum of squared differences between the observed and predicted values

5) Make predictions: Use the regression line equation to make predictions for dependent variable (Y) given new values of the independent variable (X) for example, if you have equation

$$Y = 1.30x +$$

Consider ~~price~~ predicting the price of house based on Square foot

Suppose your data shows that for every additional 500sq. feet price increases by \$50,000

Your regression equation might be

$$\text{price} = 100,000 + 200 \times \text{Square foot}$$

Here \$100,000 is the base intercept (base price) and

\$200 per square foot is the slope

If you want to predict the price of house with 2,500 square feet

$$\text{price} = 100,000 + 200 \times 2500$$

$$= 600,000$$

$$\text{predicted price} = 600,000$$